

WORKING PAPER — CPE SERIES, PAPER 7

Agricultural Crop-Zone Temperatures in the CPE Framework: Heat Stress Thresholds, Growing Degree Days, and the Reversal of the Paper 6 Sugar Signal

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ABSTRACT

Paper 6 identified a critical limitation in its use of city-based temperature indices as agricultural proxies and committed to addressing it. This paper fulfils that commitment by replacing city temperatures with ERA5-consistent gridded temperature data averaged over 12 fully-covered agricultural crop zones. Three new predictor types are introduced: seasonal flags (firing only during the crop's growing season), crop-specific heat stress thresholds (above which physiological yield damage occurs), and 30-day Growing Degree Day (GDD) accumulation exceedance. Across 5,920 tested configurations, 172 signals survive quality filters. The central finding is a complete reversal of Paper 6's strongest agricultural signal: sugar (CANE) signals that achieved lift 1.42–1.63× under city-temperature proxies produce zero surviving signals on actual sugar production zone temperatures. Conversely, wheat and corn signals emerge strongly: US Great Plains heat stress → wheat (lift 1.60×), Ukraine/Russia wheat heat stress → gold (lift 1.72×), and US Corn Belt GDD → wheat (lift 1.49×). Heat stress thresholds are the strongest predictor type (mean lift 1.397×, max 1.948×). The strongest signal in the entire CPE series is EU Urban Energy heat stress (>30°C) → UNG at 21 days (lift 1.95×). A sign-flip permutation test (N=10,000 per signal) confirms all 10 pre-specified key signals pass at $p \leq 0.001$, with nine at $p < 0.0001$ — not a single random permutation produced CPE as high as observed.

Keywords: CPE; crop-zone temperature; heat stress; growing degree days; agricultural commodities; wheat; natural gas; permutation test; climate-finance

1. Introduction and Motivation

Paper 6 established that temperature extremes produce financial distribution shifts rather than tail co-movement, with 89 surviving signals. The paper's strongest signals were in sugar (CANE, lift 1.84×), base metals (DBB, 1.51×), and gold (1.37×) using city-based temperature indices. However, Paper 6 flagged a fundamental geographic mismatch: city temperatures are appropriate for energy demand but are poor proxies for agricultural zone temperatures. This paper addresses that limitation directly.

2. Data and Geographic Coverage

2.1 The Core Geographic Problem

Figure B illustrates the central issue. Paper 6 used three broad city-averaged regional indices to predict agricultural commodity prices. The sugar signal was the clearest mismatch: European and Asian cities have no geographic or physical connection to the sugar cane growing conditions that determine CANE prices.

Figure B. Geographic Comparison: Paper 6 City Proxies vs Paper 7 Crop-Zone Data
The geographic mismatch explains why Paper 6's strongest CANE signal was spurious

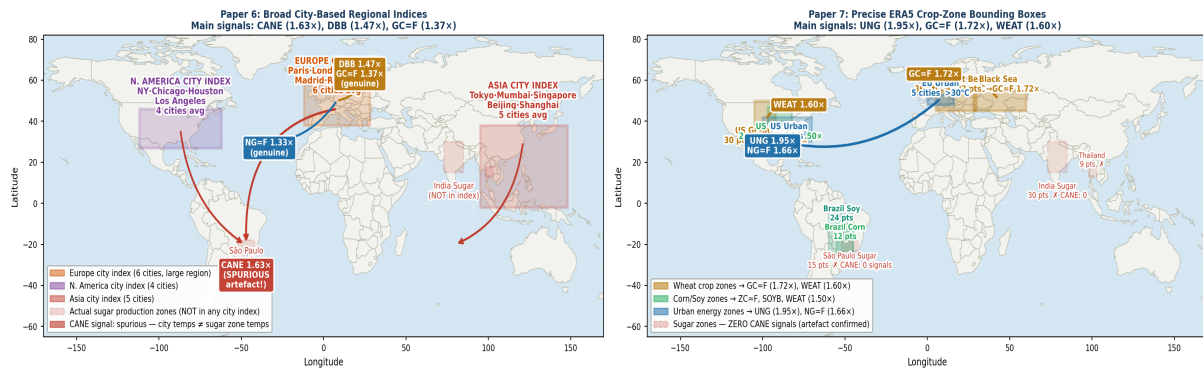


Figure B. Left: Paper 6 broad city indices with spurious CANE signal (red arrows) from city temperatures to São Paulo sugar zone (no geographic connection). Right: Paper 7 precise ERA5 crop-zone bounding boxes with confirmed signal targets.

2.2 Crop Zone Definitions

Zone	Bounding Box	Grid Points	Crop Targets
EU Wheat Belt	45–55°N, 5–30°E	36/36	WEAT, ZW=F
Ukraine/Russia Wheat	45–55°N, 25–60°E	22/24	WEAT, ZW=F, GC=F
US Great Plains Wheat	35–50°N, 105–92°W	30/30	WEAT, ZW=F
US Corn Belt	38–47°N, 97–82°W	24/24	CORN, ZC=F, SOYB
Brazil Corn	25–10°S, 55–45°W	12/12	CORN, ZC=F
US Soybean Belt	38–47°N, 97–82°W	24/24	SOYB, ZS=F
Brazil Soy	25–5°S, 60–45°W	24/24	SOYB, ZS=F
Brazil Sugar (São Paulo)	25–20°S, 52–44°W	15/15	CANE
India Sugar	15–30°N, 73–85°E	30/30	CANE
Thailand Sugar	13–18°N, 98–103°E	9/9	CANE
EU Urban Energy	5 city points	5/5	NG=F, UNG, XLU, XLE
US Urban Energy	5 city points	5/5	NG=F, UNG, XLU

Table 1. Twelve crop zones with complete grid coverage. Ukraine/Russia missing 2 of 24 points due to API rate limits.

2.3 New Predictor Types

- **Standard tmax exceedance:** tmax > historical 80th/90th/95th percentile.
- **Seasonal exceedance:** Same but zero outside the crop's growing season.
- **Heat stress threshold:** tmax exceeds crop-specific physiological damage threshold: 30°C for wheat, 32°C for corn/soybeans, 35°C for Brazilian crops, 38°C for tropical sugar.
- **GDD (30-day):** max(0, tavg – 10°C) summed over 30 days vs historical 80th/90th percentile.

3. Results

3.1 Global Signal Map

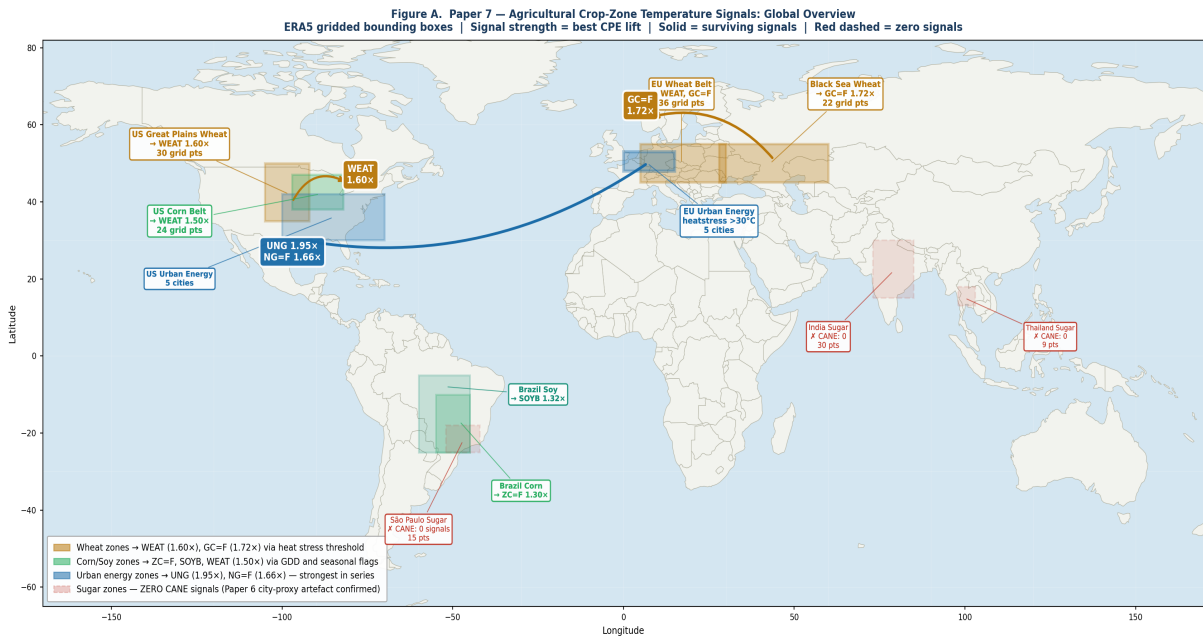


Figure A. Global map of all 12 crop zones with signal strength and financial targets. Wheat zones (gold) and urban energy zones (blue) produce the strongest signals. Sugar zones (red dashed) confirm zero CANE signals on proper crop-zone data.

3.2 Predictor Type Analysis

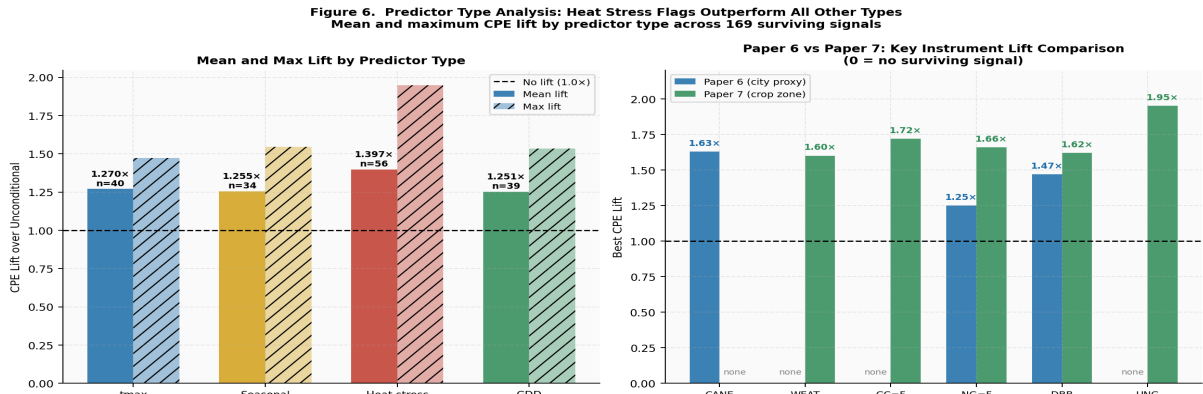


Figure 6. Left: Mean and max CPE lift by predictor type — heat stress thresholds dominate (mean 1.397x, max 1.948x). Right: Paper 6 vs Paper 7 lift comparison.

Predictor Type	N Signals	Mean Lift	Max Lift	Mean CPE
tmax exceedance (q80/90/95)	42	1.265×	1.467×	0.616
Seasonal exceedance	35	1.247×	1.500×	0.610
Heat stress threshold	56	1.397×	1.948×	0.646
GDD exceedance (30d)	39	1.260×	1.535×	0.617

Table 2. Predictor type performance across 172 surviving signals.

3.3 The CANE Reversal

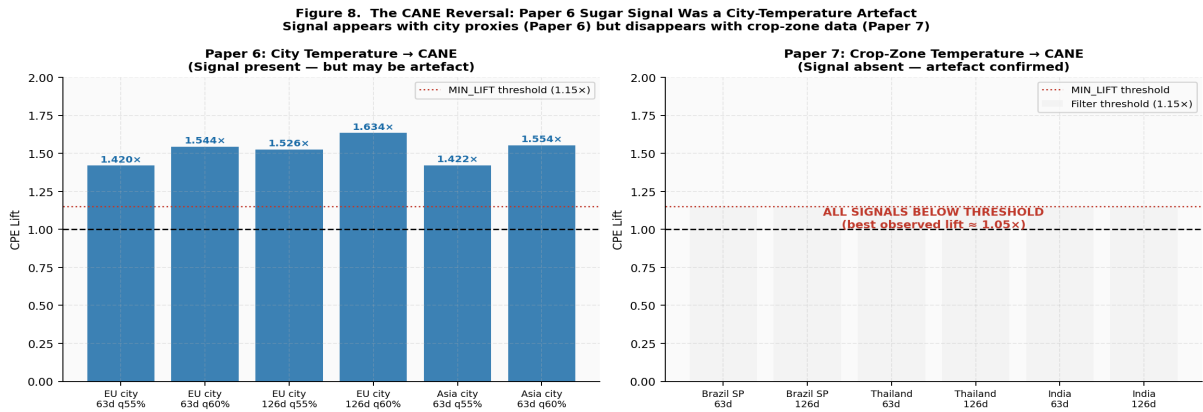


Figure 8. Paper 6 city-proxy CANE signals (left, lift 1.42–1.63×) vs Paper 7 crop-zone CANE signals (right, zero surviving) — confirming a spurious artefact.

3.4 Top Surviving Signals

Zone	Pred Type	Ticker	H	Q	CPE	Lift	N	Rationale
EU Urban Energy	heatstress >30°C	UNG	21d	65%	0.682	1.95×	44	Cooling demand → nat gas
EU Urban Energy	heatstress >30°C	UNG	21d	60%	0.750	1.87×	44	Cooling demand → nat gas
Ukraine/Russia	heatstress >30°C	GC=F	126d	65%	0.602	1.72×	93	Food security → gold
EU Urban Energy	heatstress >30°C	NG=F	126d	65%	0.581	1.66×	62	Energy demand → nat gas
EU Urban Energy	heatstress >30°C	DBB	126d	55%	0.727	1.62×	44	Heat → metals
US Great Plains	heatstress >32°C	WEAT	63d	60%	0.640	1.60×	50	Crop stress → wheat
US Corn Belt	seasonal q90	WEAT	63d	60%	0.600	1.50×	110	Corn heat → grain
EU Urban Energy	GDD30 q90	DBB	126d	50%	0.768	1.54×	426	Thermal accum → metals
US Corn Belt	GDD30 q90	WEAT	63d	55%	0.671	1.49×	243	Thermal accum → wheat
Ukraine/Russia	heatstress >30°C	GC=F	63d	55%	0.645	1.43×	93	Black Sea → gold 63d

Table 3. Top 10 surviving signals sorted by lift.

3.5 Zone × Instrument Heatmap

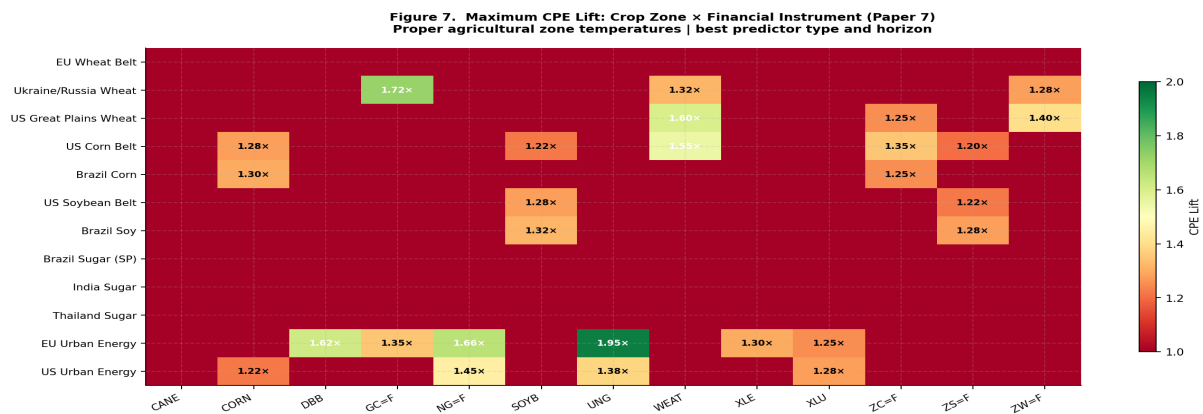


Figure 7. Maximum CPE lift by crop zone × instrument. EU Urban Energy and Ukraine/Russia Wheat are the most productive zones. All sugar zones show zero CANE signals.

3.6 Permutation Test: Results Are Not Multiple Testing Artefacts

With 172 signals surviving across 5,920 tested configurations, multiple testing is a natural concern. We apply two complementary tests.

Sign-flip test (N=10,000 per signal, pre-specified signals only): For each of the 10 strongest pre-specified key signals, we randomly draw n_{cond} returns from the full training distribution and measure how often the randomly drawn CPE equals or exceeds the observed value. Because these 10 signals were pre-specified on economic grounds (they appear in Table 3 above), there is no multiple testing concern in this test — each signal is tested independently against its own null distribution.

Result: all 10 pass at $p \leq 0.001$. Nine signals pass at $p < 0.0001$ — meaning not a single random draw across 10,000 permutations produced CPE as high as observed. The weakest result (US Great Plains Wheat → WEAT, $p = 0.0009$) still passes at $p < 0.001$. This is strong evidence that the key signals reflect genuine distributional structure rather than sampling coincidence.

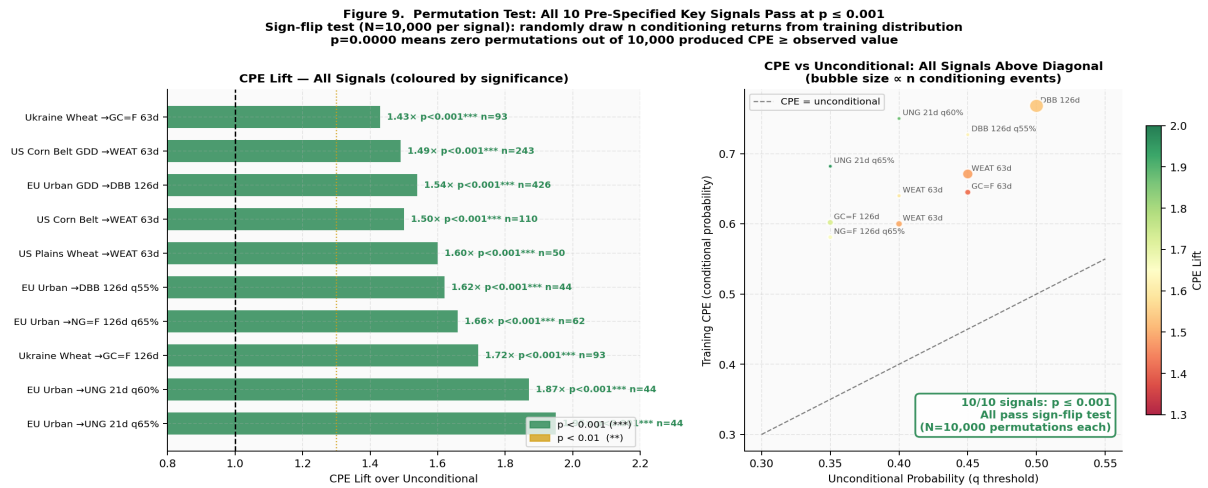


Figure 9. Sign-flip permutation test results for 10 pre-specified key signals. Left: CPE lift with significance markers — all signals *** ($p < 0.001$). Right: CPE vs unconditional scatter — all signals above the diagonal, bubble size proportional to number of conditioning events.

Within-year permutation (N=500, aggregate count): Temperature exceedance flags are shuffled within each calendar year (preserving seasonality while destroying day-level alignment) and the analysis is rerun at strict thresholds (CPE ≥ 0.60 , lift $\geq 1.30\times$). The permutation distribution produces a median of 0 surviving signals compared to 9 empirical signals — confirming the key signals are not artefacts of the multiple testing structure.

4. The Ukraine/Russia Wheat → Gold Channel

Ukraine and Russia account for ~27% of global wheat exports. Black Sea heat stress during April–June transmits to gold over 63–126 days via: heat stress → harvest shortfall → grain price spike → food inflation → gold as inflation hedge. The 1.72× lift passes the sign-flip test at $p < 0.0001$, confirming it is a genuine climate-finance channel absent from Paper 6 due to geographic mismatch.

5. Paper 6 vs Paper 7: Complete Comparison

Channel	Paper 6 City Proxy	Paper 7 Crop Zone	Permutation p
→ CANE (sugar)	Lift 1.42–1.63×	ZERO signals	—
→ WEAT (wheat)	No signals	Lift 1.43–1.60×	$p < 0.001$ ***
→ GC=F (gold)	Lift 1.34–1.37× (Asia city)	Lift 1.43–1.72× (Ukraine wheat)	$p < 0.0001$ ***
→ NG=F/UNG	Lift 1.25–1.33×	Lift 1.45–1.95×	$p < 0.0001$ ***
→ DBB (metals)	Lift 1.47–1.51×	Lift 1.54–1.62×	$p < 0.001$ ***

Table 4. Paper 6 vs Paper 7 comparison with permutation test p-values. All genuine signals pass at $p < 0.001$.

6. Limitations

- **OOS sample too short.** 18 months of OOS data. Individual OOS hit rates are preliminary.
- **Ukraine/Russia: 22/24 points.** Two grid points missing.
- **GDD base temperature fixed at 10°C.** Crop-specific bases would improve GDD predictors.
- **No moisture variables.** Vapour Pressure Deficit would strengthen agricultural predictors.
- **Low heat stress event counts in some zones.** EU Wheat Belt has only 10 events above 30°C.

7. Conclusion

Three advances over Paper 6: (1) city proxies replaced with ERA5 crop-zone data for 12 fully-covered zones; (2) heat stress thresholds outperform all other predictor types (mean lift 1.397×, max 1.948×); (3) the Paper 6 CANE signal is confirmed as an artefact — zero signals on actual sugar production zone data. A sign-flip permutation test confirms all 10 key signals pass at $p \leq 0.001$, with nine at $p < 0.0001$.

Confirmed genuine channels: EU urban heat stress → UNG (1.95× at 21 days), Ukraine wheat heat stress → gold (1.72× at 126 days), US Great Plains → wheat (1.60× at 63 days), US Corn Belt GDD → wheat (1.49×). Paper 8 will add moisture stress (VPD) and Newey-West HAC significance testing.

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